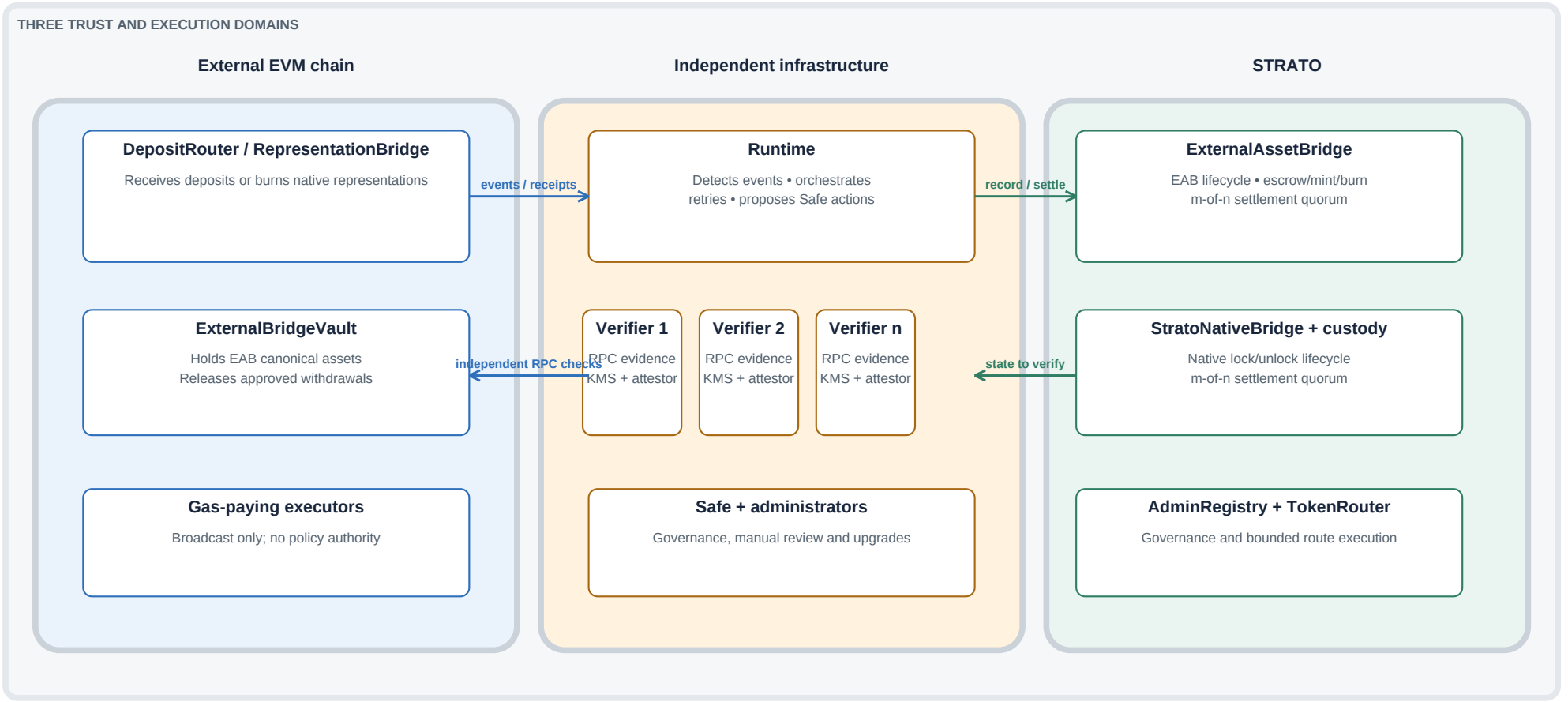
STRATO Bridge Architecture

External Asset Bridge (EAB), native bridge, verifier quorum and infrastructure interactions

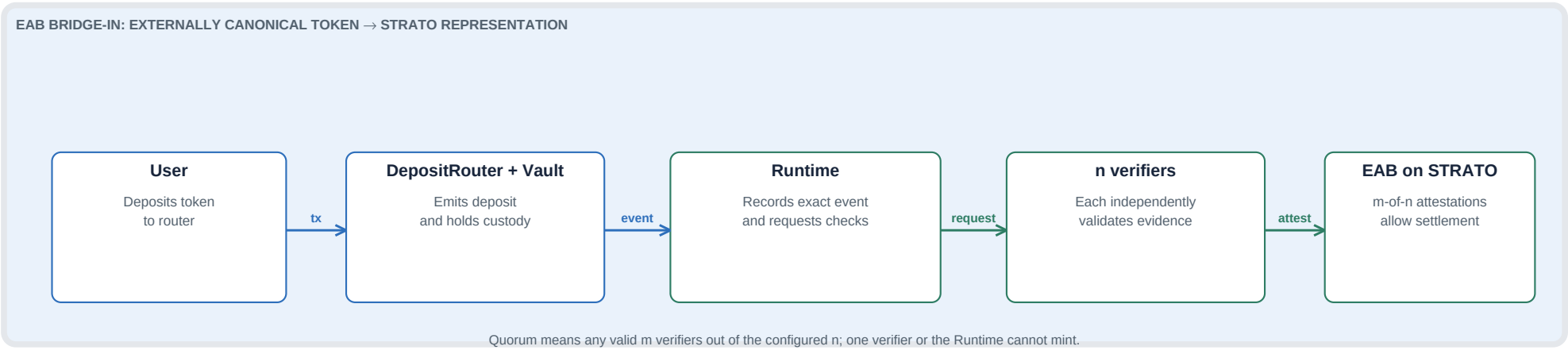


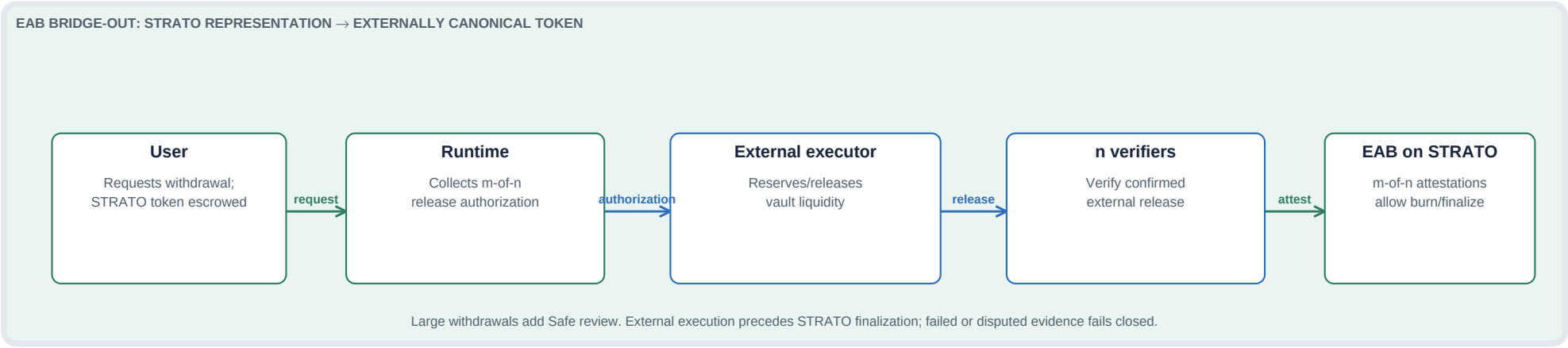
Core security model

A transfer advances only after **m valid verifiers out of n configured verifiers** independently validate the same canonical evidence. Runtime coordinates work but is not quorum. Executors pay gas but do not set policy. Safe/AdminRegistry govern exceptional and configuration changes.

1. EAB interaction model

Canonical assets remain in an external-chain vault; STRATO represents the corresponding bridged value.





Deposit invariant Vault custody and the router event must match the reviewed route, amount, sender/recipient and finalized canonical receipt.	**Withdrawal invariant** External release is confirmed before STRATO escrow is burned/finalized. Reservations and refunds remain evidence-bound.	**Governance boundary** Large or reviewed operations require Safe/AdminRegistry approval in addition to verifier quorum; governance does not replace evidence.

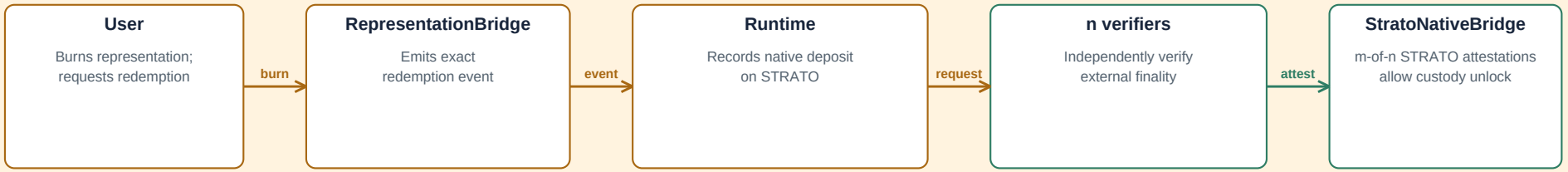
2. Native interaction model

STRATO-native value is locked in STRATO custody and represented by a mintable/burnable token on the external chain.

NATIVE BRIDGE-OUT: STRATO-NATIVE TOKEN → EXTERNAL REPRESENTATION

Destination attestation signers are KMS-backed identities; signer-set version and lane are bound into the signed message.

NATIVE BRIDGE-IN: EXTERNAL REPRESENTATION REDEMPTION → STRATO-NATIVE TOKEN

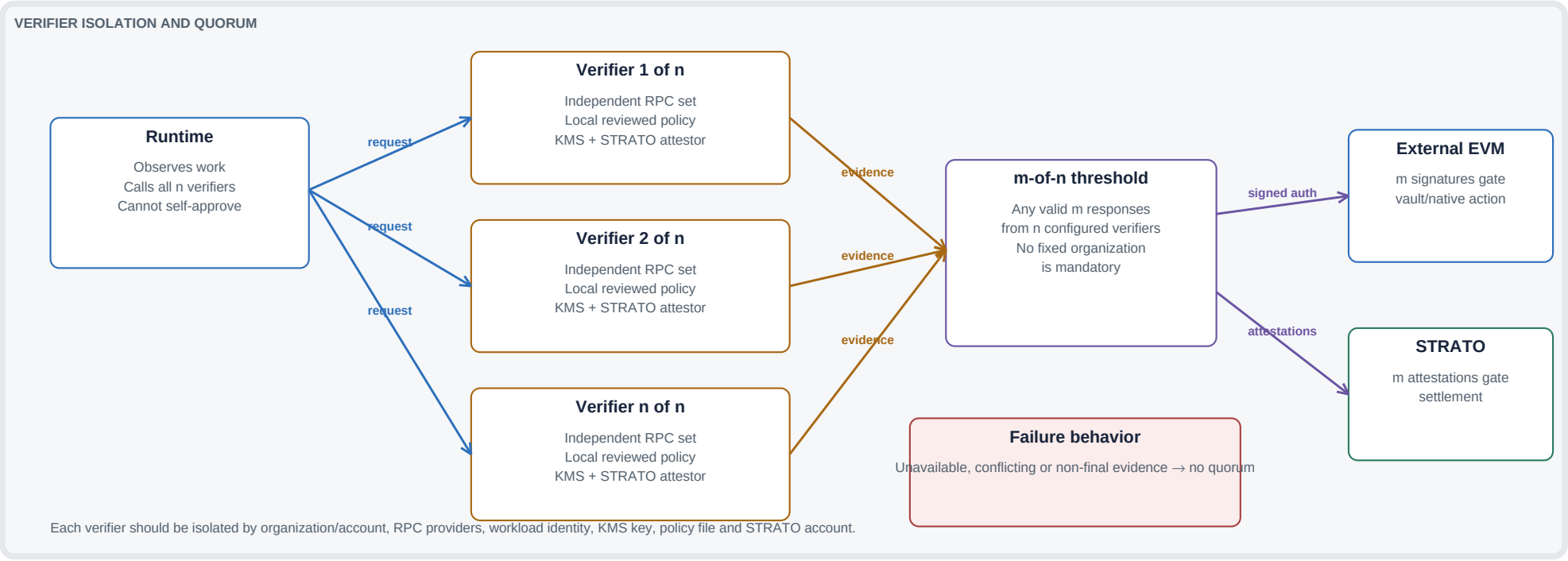


Refund and cancellation paths also require evidence-bound m-of-n verification before value is returned.

Two distinct quorums Destination minting consumes m-of-n EIP-712 signatures. STRATO custody release/finalization consumes m-of-n on-chain settlement attestations.	**Two execution lanes** An instant-path executor can submit eligible mints; the manual lane executes through the Safe. The signed message binds the selected lane.	**Replay and change protection** Messages bind source/destination identities, withdrawal or redemption identity, route, amount and signer-set version.

3. m-of-n verifiers and infrastructure

Each verifier is an independent security principal—not merely another process behind the Runtime.



Actor / component	Authority and responsibility	Must not be trusted to do alone
Runtime workload	Scans chains and STRATO, records work, calls all verifier endpoints, retries safely, submits transactions and creates Safe proposals.	Cannot manufacture verifier quorum, alter verifier policy, sign as a Safe owner or treat its own RPC view as final evidence.
Each verifier workload	Checks exact receipt/event, canonical block, finalized head, confirmation floor and local policy using independent RPC evidence.	Cannot settle alone when m > 1; cannot rely on Runtime-provided evidence without independently validating it.
Verifier KMS identity	Signs EAB vault authorizations or native destination attestations. The expected signer address is configured on-chain and in reviewed policy.	Cannot administer routes, upgrade contracts or spend from the Safe.
Verifier STRATO attestor	Submits that verifier's evidence-bound settlement attestation to EAB or StratoNativeBridge.	Cannot finalize without m-of-n on-chain attestations and matching recorded evidence.

4. Actors, controls and operating boundaries

The architecture separates observation, authorization, execution, settlement and governance.

Actor / component	Authority and responsibility	Must not be trusted to do alone
User	Initiates deposits, withdrawals or native redemptions and receives the destination asset.	Cannot bypass route policy, finality, custody or quorum.
External-chain contracts	DepositRouter records EAB deposits; ExternalBridgeVault holds/releases canonical assets; RepresentationBridge mints/burns native representations.	Do not decide STRATO settlement without configured cryptographic or on-chain quorum.
External gas executor	Broadcasts EAB vault or native instant-path transactions using a funded KMS-backed account.	Has no route-admin, Safe-owner or unilateral mint/release authority.
STRATO bridge operator	Records evidence and advances permitted lifecycle steps after required checks.	Cannot substitute for m-of-n settlement verifier attestations.
Relayer	Submits permissionless or narrowly authorized finalization after quorum exists.	Cannot create evidence, approve reviews or alter bridge policy.
Safe owners / STRATO admins	Configure contracts, approve reviewed operations, execute manual native mints and govern upgrades.	Governance approval does not prove external execution or replace verifier quorum.
TokenRouter	Executes bounded post-deposit routing when the reviewed action and dependencies permit it.	Cannot mint beyond EAB settlement or route into unapproved dependencies.
Custody contracts	ExternalBridgeVault holds EAB canonical assets; StratoNativeCustodyVault holds STRATO-native assets.	Release only through their configured bridge and authorization rules.

Required infrastructure separation

Run one Runtime per network responsibility and n isolated verifier stacks. Give every verifier distinct workload identity/KMS key, STRATO attestor account, reviewed policy and independent RPC evidence. Persist Runtime cursors and Safe proposal journals. Fund only transaction-submitting accounts: external executors with native gas and STRATO operator/attestors/admins with vouchers.

Terminology: “m-of-n” means any m valid, configured verifier identities out of n total identities. The actual values are deployment configuration—not an architectural constant. EAB settlement, EAB release authorization, native destination mint authorization and native STRATO settlement may use separate signer sets and thresholds even when implemented by the same verifier organizations.